

Module - 4

Q1- Given a positive integer N , print all the no. from 1 to N in increasing order, separated by a single space. Your task is to write a program that prints the required sequence. After completing the program, analyze your solⁿ by calculating its TC & SC, explain why your answer is correct.

Solⁿ ① Approach:-

- Read the value of N from the user.
- Start a loop from 1 & continue until N .
- Print the current value of the loop variable in each iteration.
- Increase the loop variable by 1 after every iteration.
- The loop ends all no. from 1 to N have been printed.

TC:- $O(N)$

SC:- $O(1)$

Explanation:-

- The loop runs exactly N times.
- Each iteration performs only one print operation, which takes constant time.

- PAGE NO.
 DATE
- No extra data structure is used, so only constant extra memory is required.

Optimized approach:-

No optimized approach exists because every no. from 1 to N must be printed at least once. Therefore, $O(N)$ is the optimal time complexity.

Q:- Given a positive integer N, find the sum of the first N natural no. using a loop. Your task is to WAP that calculates & prints the required sum.

After completing the program, analyze your sol. by calculating its TC & SC, & explain your answer.

sol:- Q:- Approach:-

- Read the value of N.
- Initialize a variable $sum = 0$.
- Traverse from 1 to N using loop.
- Add the current no. to sum in every iteration.
- After the loop ends, prints the value of sum.

T.C:- $O(N)$

S.C:- $O(1)$

Explanation:-

- The loop executes N times.

- Every iteration performs one addⁿ operation.
- Only one extra variable (sum) is used, so the SC remains constant.

Optimized approach:

Instead of using a loop, the sum can be calculated using the formula:

$$\text{sum} = N * (N+1) / 2$$

- Optimized TC: $O(1)$
- Optimized SC: $O(1)$

Q3 Given two integers A & B, calculate the value of $A ^B$ (A raised to the power B) using a loop.

Your task is to write a program that computes the result & prints it.

Once your program is complete, analyze your solⁿ by calculating its TC & SC, & explain why your answer is correct.

Sol: 3 Approach:-

- Read the values of A (base) & B (exponents).
- Initialize result = 1.
- Run a loop exactly B times.
- Multiply result by A in every iteration.
- After the loop completes, print the value of result.

TC :- $O(B)$

SC :- $O(1)$

Explanation:-

- The loop runs exactly B times.
- Each iteration performs one multiplication.
- Only one extra variable (result) is used, so the extra space remains constant.

Optimized approach:-

Use Binary Exponentiation (Fast Exponentiation).

- Repeatedly square the base & reduce the exponent by half.
- This reduces the no. of multiplications significantly.

Optimized TC :- $O(\log B)$

Optimized SC :- $O(1)$